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1. Welcome

Welcome to the SEA.AI Sentry. SEA.AI systems utilize an advanced artificial intelligence engine and high quality thermal and optical cameras to provide early detection and identification of floating objects on the water – during the day and at night.

This includes objects that might escape conventional systems such as radar or AIS, including: unidentified vessels, floating obstacles, containers, buoys, inflatables, kayaks and persons in the water.

The SEA.AI system's artificial intelligence leverages the latest Machine Vision technology, best-in-class deep learning capabilities, and a constantly evolving, proprietary database of millions of annotated marine objects.



Thank you for choosing SEA.AI.

Sentry's comprehensive optical and thermal camera setup, complemented by a powerful internal processing unit, provides a higher detection range, wider detection angle and higher detection frequency, improving performance in unfavorable sea conditions.

Sentry provides you with the ability to pan 360° and tilt up to ±20° for your safety and surveillance both at anchorage and underway.

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2. Specifications

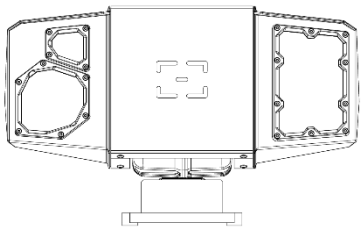
SEA.AI combines high-tech optical sensors with the latest technologies in the field of computer vision and artificial intelligence to enhance safety at sea.

SEA.AI's proprietary database is the core of the system software and its integrated AI-powered object detection and recognition:

- Over 24 millions annotated marine objects (thermal images and RGB images)
- SEA.AI's proprietary Thermal Convolutional Neural Network (CNN).
- Continuously growing and improving (new object classes, e.g., icebergs, whales; new geographic regions).
- Regular software updates will ensure best performance

What Sentry offers

- Cameras with a machine vision processor. A combination of high-resolution low-contrast and thermal cameras.
- Detection of temperature differences up to 0,03°C.
- Pan and Tilt unit: 360° / ±20°.
- Mechanical and Digital image Stabilization.
- Operation Modes. (Manual Control, Collision Avoidance Mode, Surveillance Mode)
- AI based Target-Tracking.



Ultimate performance with combination of 4 cameras:

Thermal Camera, Long Range	LWIR 640x512px, 8° HFoV, 55mm lens
Thermal Camera, Close Range	LWIR 640x512px, 32° HFoV, 14mm lens
RGB Camera, Long Range	2560x1944px, 9,3° HFoV, 35mm lens
RGB Camera, Close Range	2560x1944px, 39,2° HFoV, 8mm lens

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3. Installation

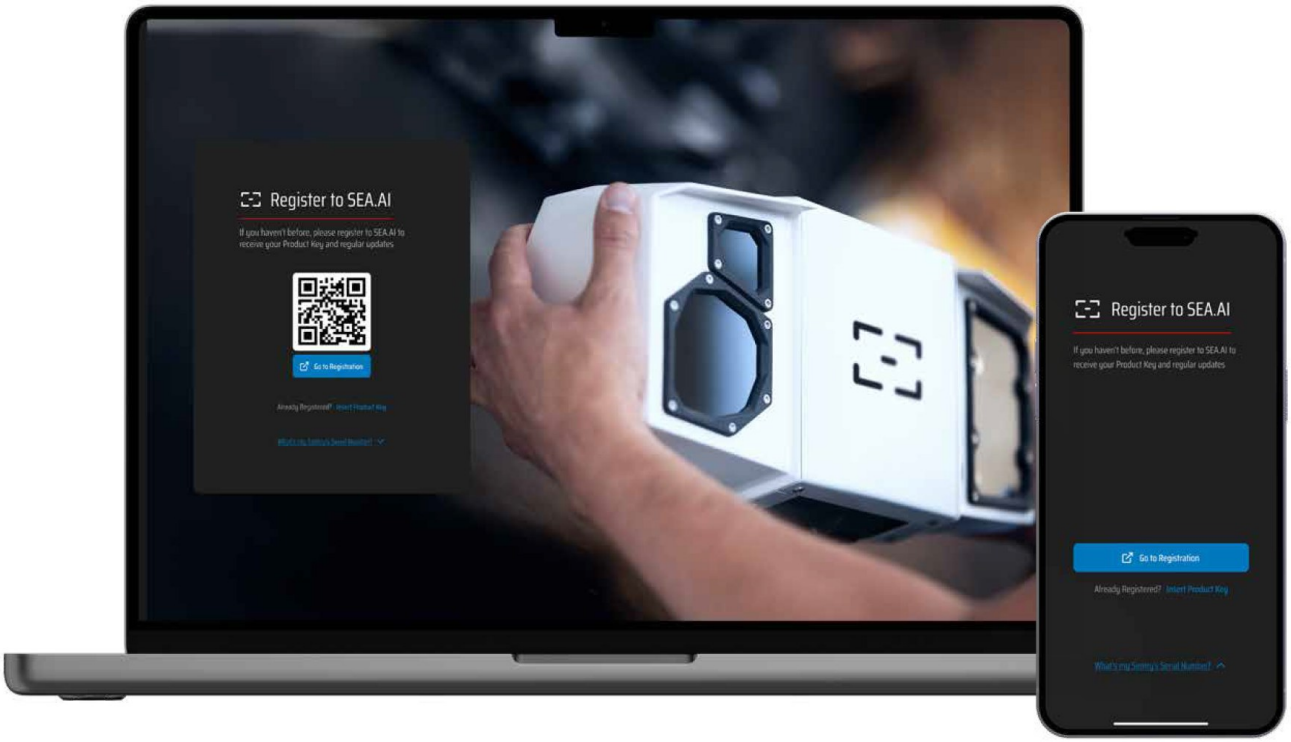
Note: Installation must be completed by a certified SEA.AI installer. Please check the installation guide in the SEA.AI document library. Accessible under www.sea.ai/document-library/ in the section Sentry.

4. Operating Instructions

Product registration and activation:

To ensure your new SEA.AI Sentry delivers optimal performance and stays up to date with the latest advancements in Artificial Intelligence, periodic updates are essential.
To receive Software Updates and your product key to activate your SEA.AI Sentry you need to register your product.

- Register your product easily by logging into your SEA.AI Connect account at: https://connect.sea.ai/account/login
- If you do not have an account yet, you can simply create one at: https://connect.sea.ai/account/register



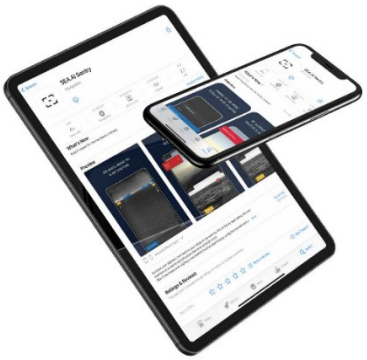
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Download SEA.AI Sentry app:

- How to connect your mobile devices (phone, tablet)?
    - Download the latest version of the SEA.AI App.

App Store:

Google Play Store:



    - Connect to the SEA.AI app using your boat WIFI via router to access the SEA.AI Sentry.
    - When connected to the Wi-Fi, the SEA.AI App will automatically find your new Sentry. Please make sure your device stays connected to the Wi-Fi.

Connecting Onboard PC:

- Download the SEA.AI Sentry Windows/Mac application from SEA.AI Document library
- Open the application. Your PC will find automatically the Sentry if it's connected to the same network.
- You can also open any Web browser and enter "sentry.local/" or the IP address set during installation in the access bar.

Please note: The application supports simultaneous usage for up to two devices. If a third device attempts to connect to Sentry, there will be an option to replace one of the devices already connected.

Start up:

- Switch on your device and the Sentry unit. Please note that the startup process for all devices may take up to 5 minutes.
- Once the devices are powered and ready, open your SEA.AI App on your mobile device, PC or MFD. If you are using a web browser Google Chrome is recommended.

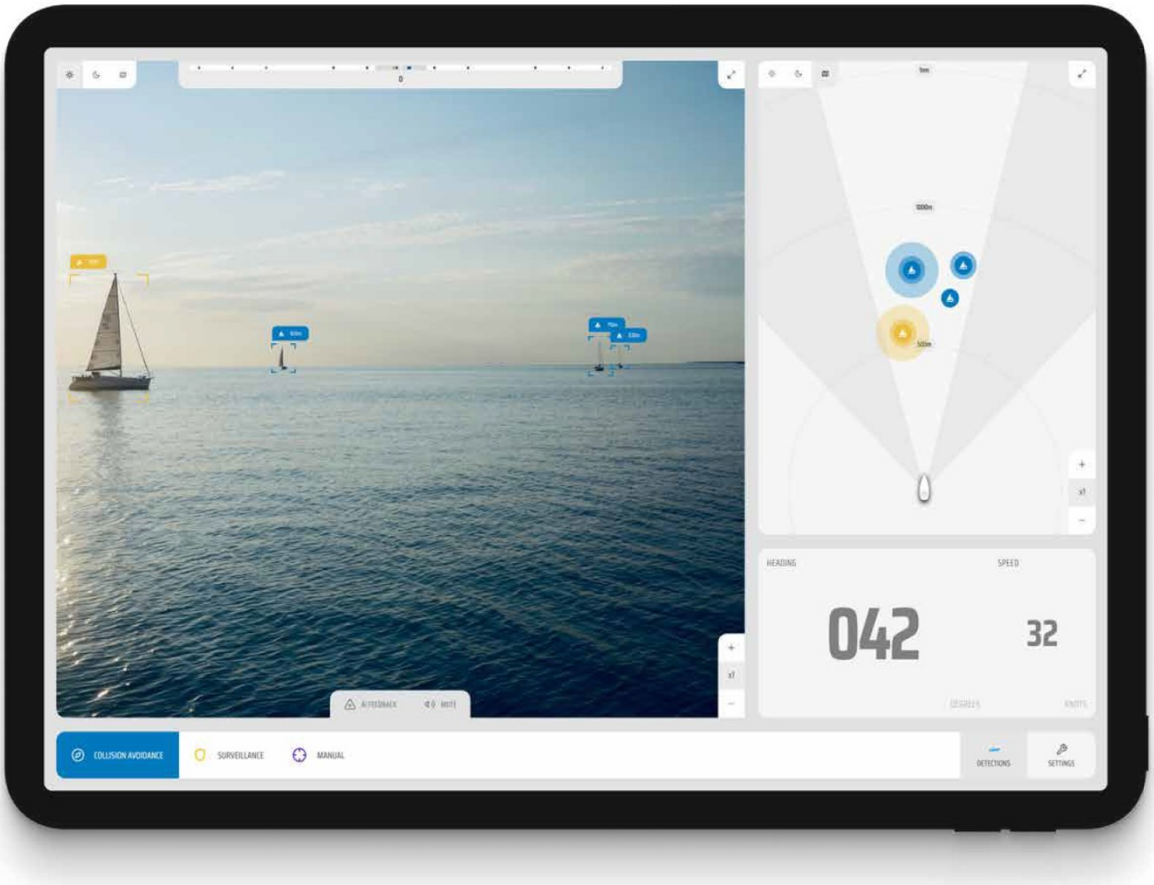
Please note : Make sure your device and SEA.AI Sentry are in the same local network!

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Interface overview:

SEA.AI systems include three main views: Colour Camera view / Thermal Camera view / Map view

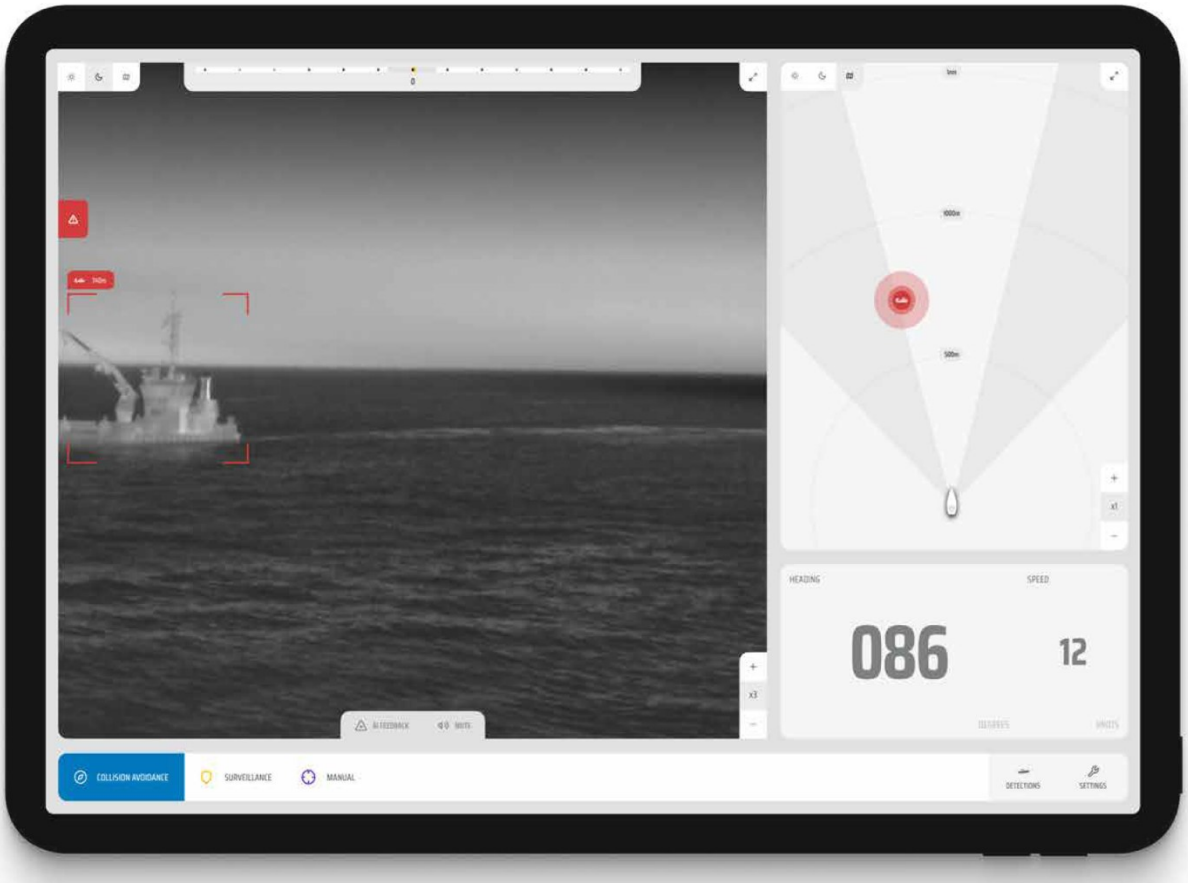
- Colour Camera view
    - In this view mode, the objects detected by the thermal camera will be visually highlighted with an overlayed bounding box, allowing for easy identification of their presence and location. The colour view provides real-time images captured by the colour camera, ensuring seamless functionality in both daylight and low-light scenarios.



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- Thermal Camera view

    - The thermal view exhibits real-time images captured by the thermal camera, operating seamlessly day and night. It proves highly beneficial in scenarios with reduced visibility, including low-light conditions or situations where visual clarity is challenging. It relies on the capabilities of the thermal camera to identify such items, making it an indispensable tool in challenging visibility situations.



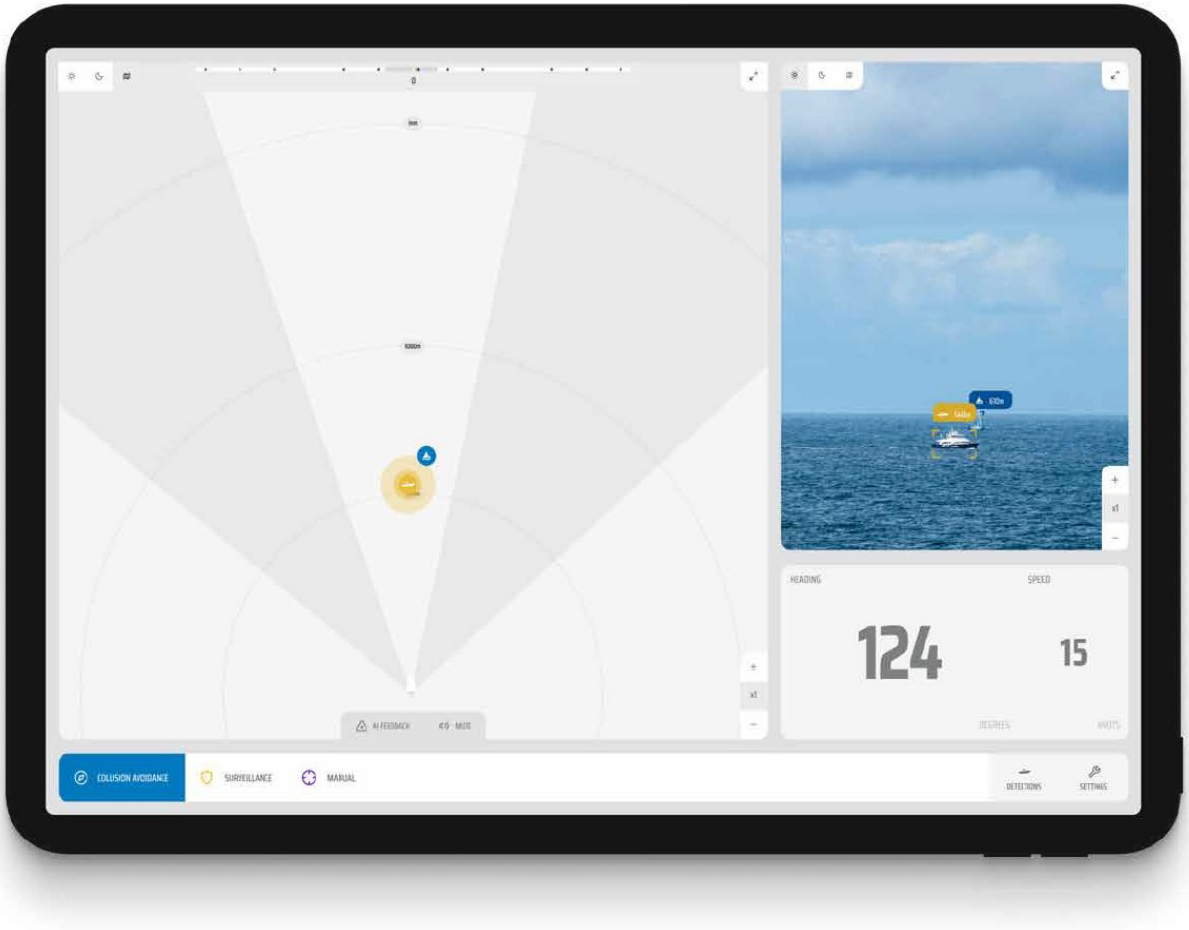
Each view will display bounding boxes around the detections.



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- Map view

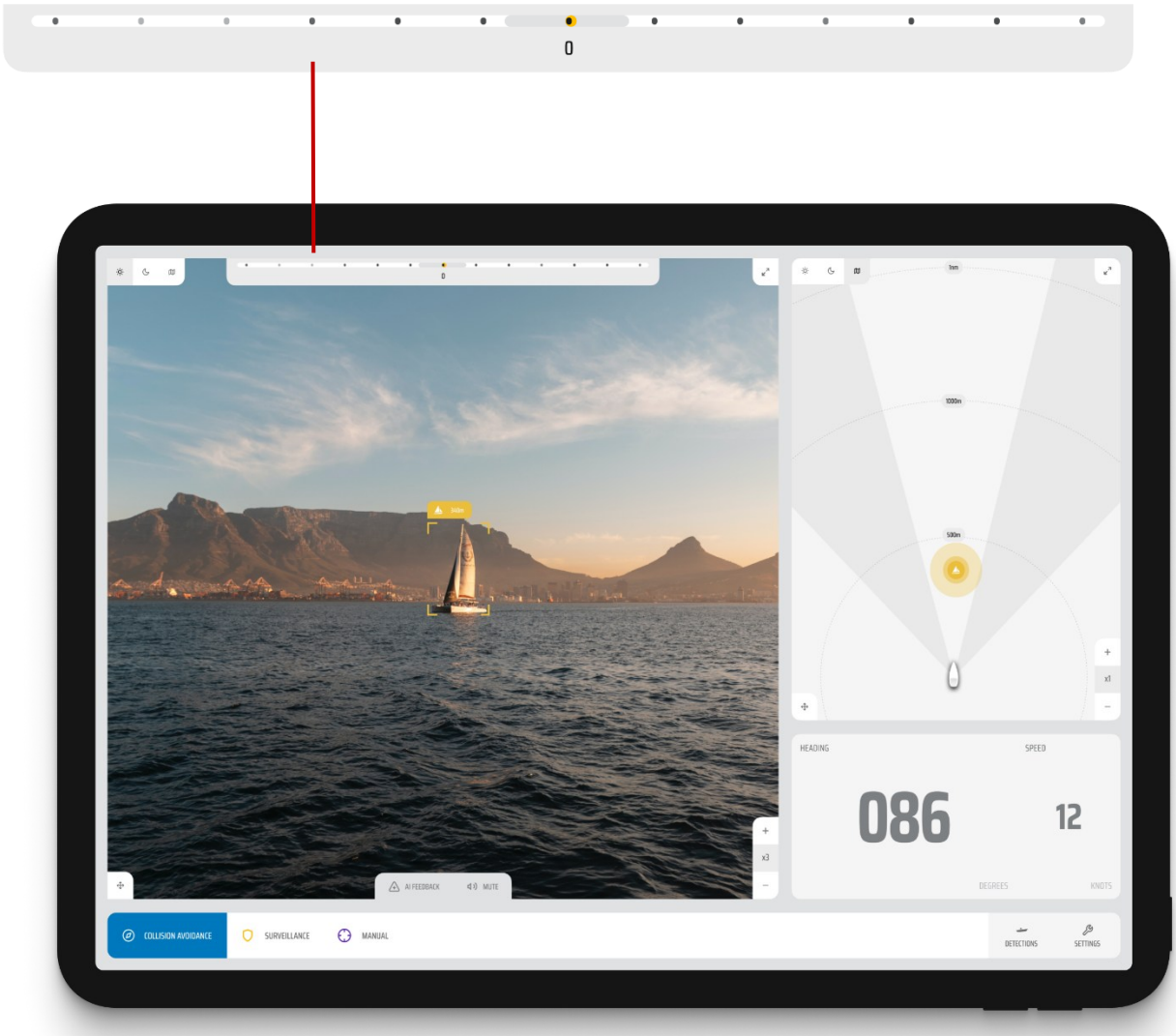
  - The map view presents a comprehensive display of all detected objects, along their corresponding threat levels (red, yellow, and blue).
    In the event that the system identifies a potentially hazardous object, an acoustic alarm is triggered.



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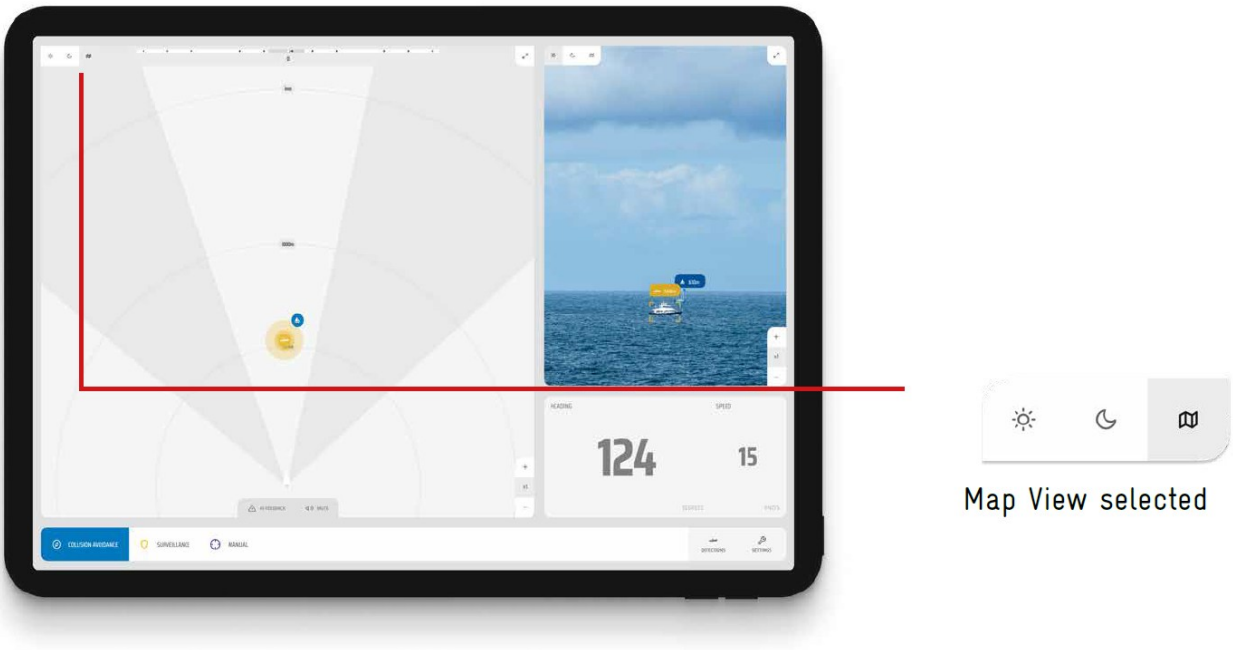
- Horizon Bar
    - At the top of the screen, the Horizon Bar indicates the camera's direction relative to the boat. By clicking on any point along the bar, you can easily select the desired camera angle, allowing for quick navigation to objects of interest. The bar also displays detected objects around the vessel with dots, enhancing your situational awareness and helping you understand your surroundings more intuitively.



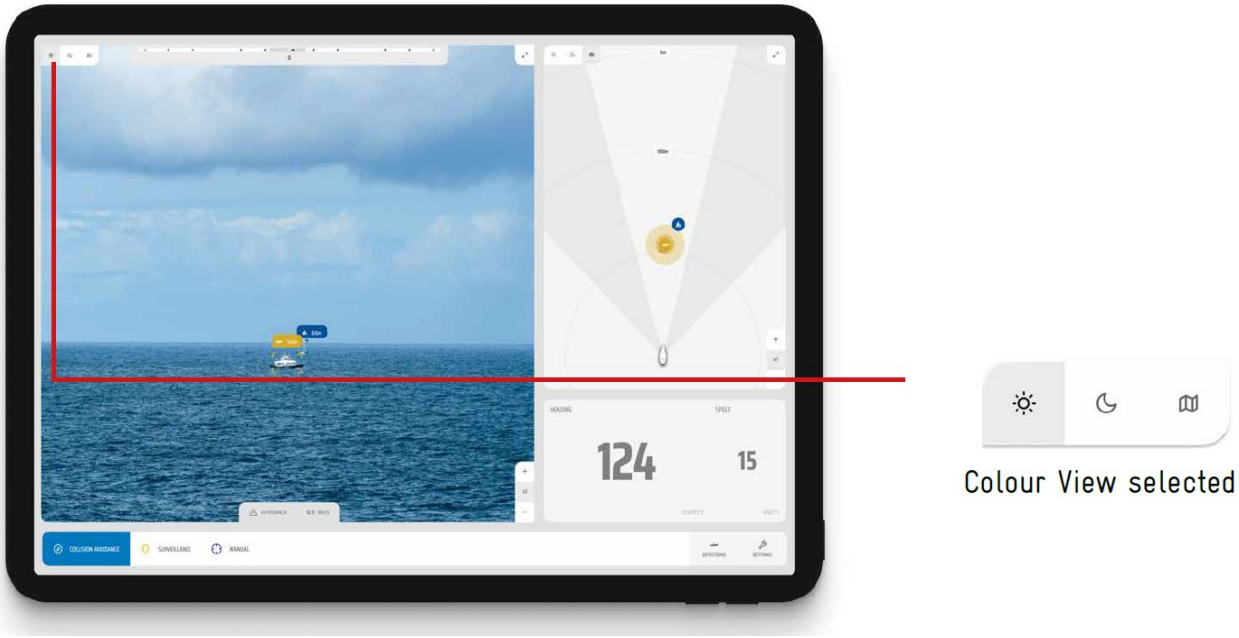
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- View switch option

    - With our double-view system, you have the flexibility to select your preferred view on the left or on the right side. You can choose between the thermal view, colour view, or the Sentry map.



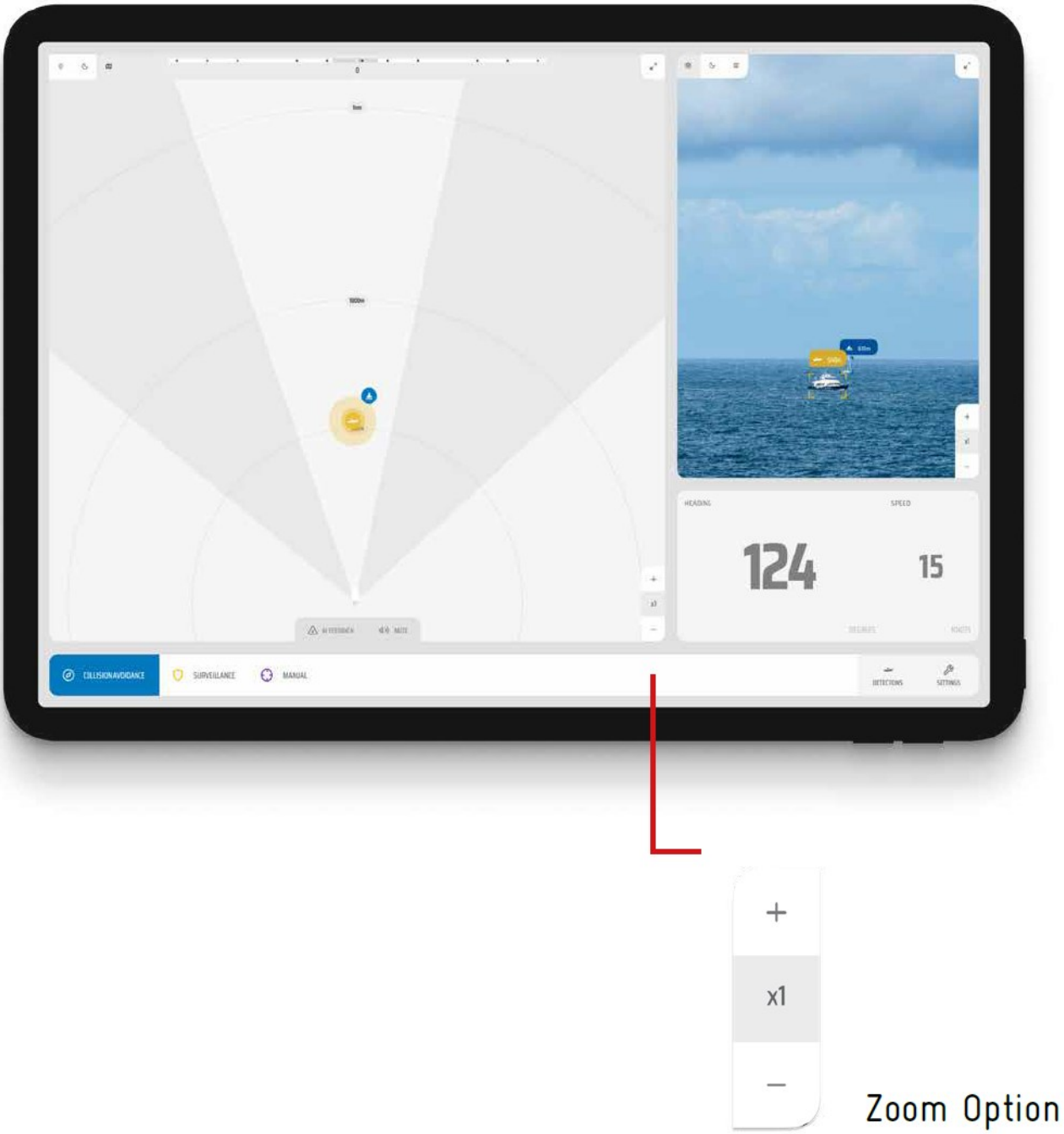
Map View selected



Colour View selected

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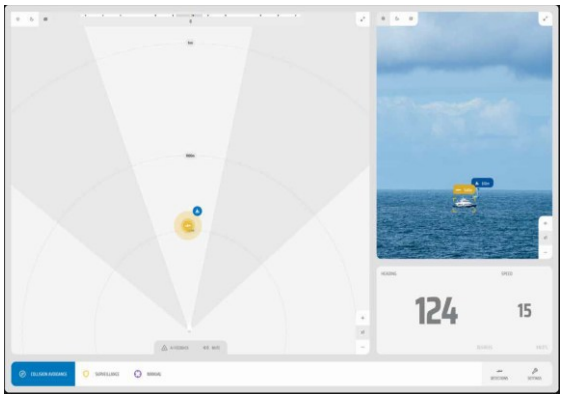
- Furthermore, you have the convenience of switching between the long-range and short-range camera views by simply using a pinch-to-zoom gesture on the screen or by using the zoom buttons provided. The system will automatically switch to the long-range camera view once you reach a certain zoom level.



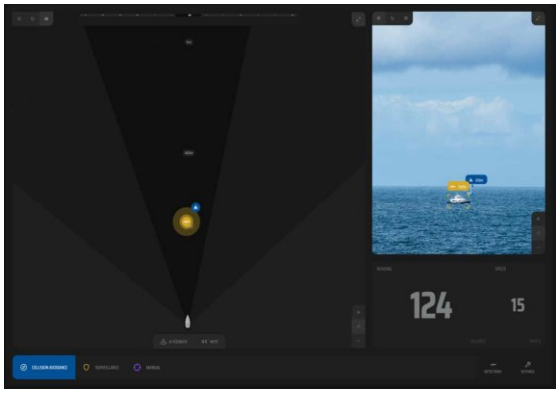
Zoom Option

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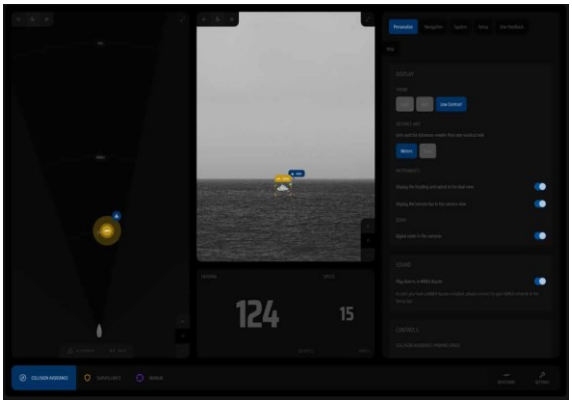
- Theme preference
    - In the Sentry app, you can choose your preferred theme:
        - Light mode: ideal for well-lit environments, ensuring easy app interaction.



        - Dark mode: A visually comfortable option for low-light conditions.



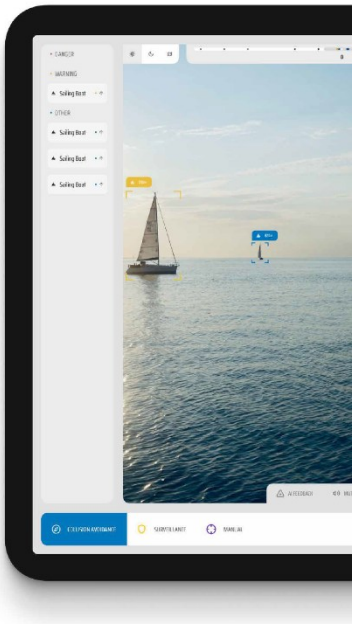
        - Night mode: Designed for use in night-time and complete darkness, minimizing screen glare.



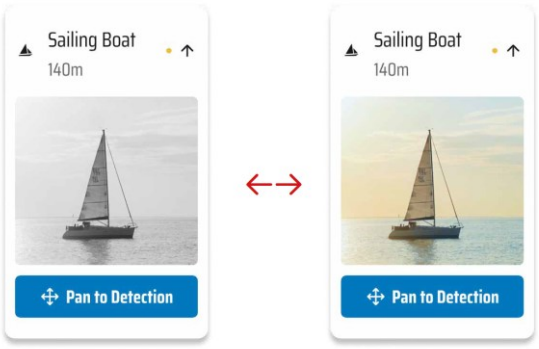
Tip: In the settings, you can also configure the opacity (brightness) of the thermal stream. To match your visibility preference and lighting conditions.

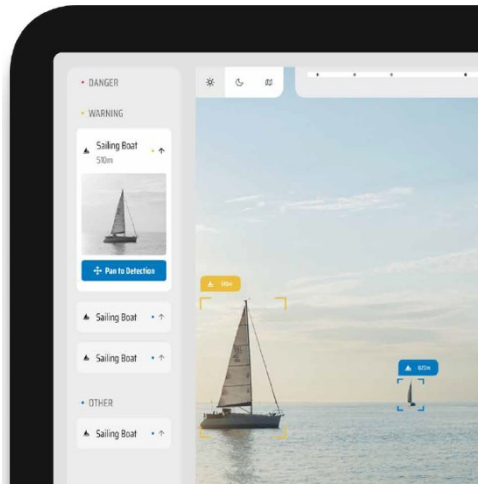
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- Detections panel



  - By clicking on the "Detections" button in the menu, the detection panel will display a comprehensive list of all detected objects, ensuring that you have access to the latest information available. The app always keeps the most recent detections conveniently sorted out by threat level. This way, you can quickly assess and prioritize them for enhanced situational awareness.
  - To view more information about a specific target, simply click on it to see additional details in the detection panel, including a detailed still image of the object last seen.
  - The app continuously updates the images of each object at any moment it is detected, showing you details on the object when last seen and keeping them current if currently detected.





  - Switching between the last seen thermal and colour image is easy: just click on the image. To ensure you have the latest information, the app always keeps the most recent detections organized based on their threat level distance. This way, you can quickly assess and prioritize them for enhanced situational awareness.

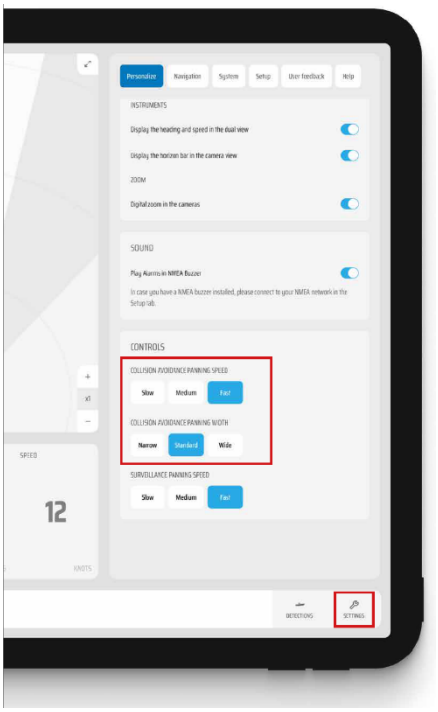
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- Operations modes

Sentry has three operation modes. Each one is customized for specific operating environments.



- Collision avoidance



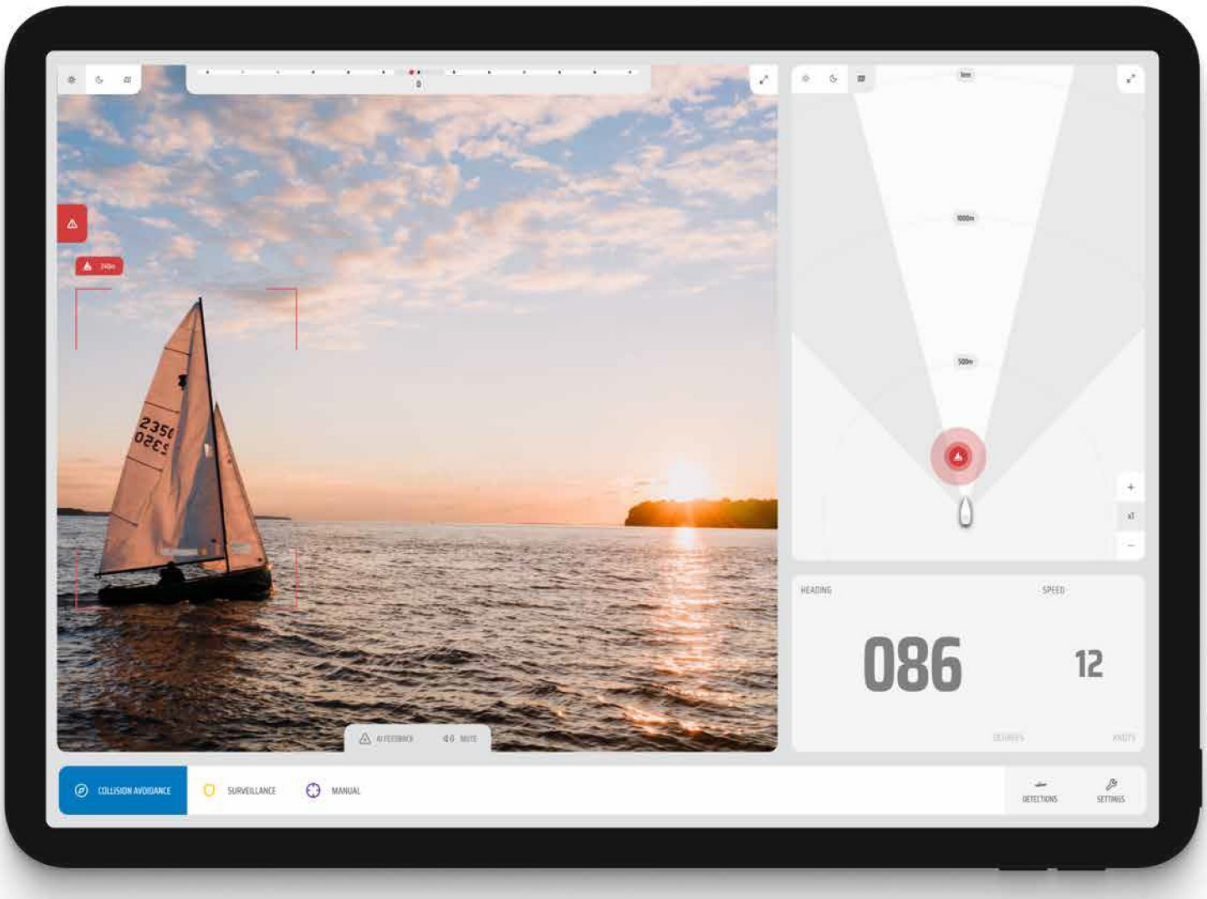
▪ In collision avoidance mode, the camera will automatically pan left and right towards the centerline of the vessel to detect any potential collision threats.

The collision avoidance mode is recommended when you are underway to help prevent accidents or collisions with automatic detection in combination with visual and acoustic alarms.

SEA.AI's smart alarms will prioritize potential hazards in 3 alarm levels based on the collision risk.



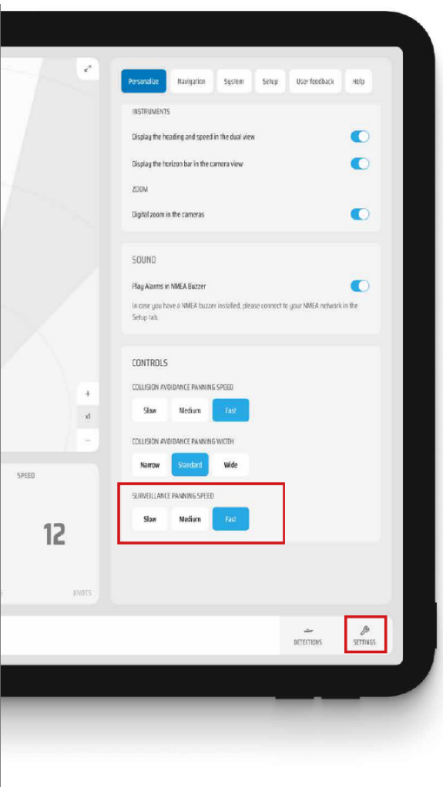
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- You have the option to customize your panning preferences in the "Personalize" tab in the settings section. The continuous scanning feature ensures timely detection of obstacles, while the customizable panning width and speed options provide flexibility and adaptability. You can choose the speed and width of the panning motion according to your preference. To lock the camera straight ahead (no panning), select the Forward View.

  Please Note: In the fast mode, the detection range is reduced due to motion blur that occurs with long-range narrow field of view thermal camera.

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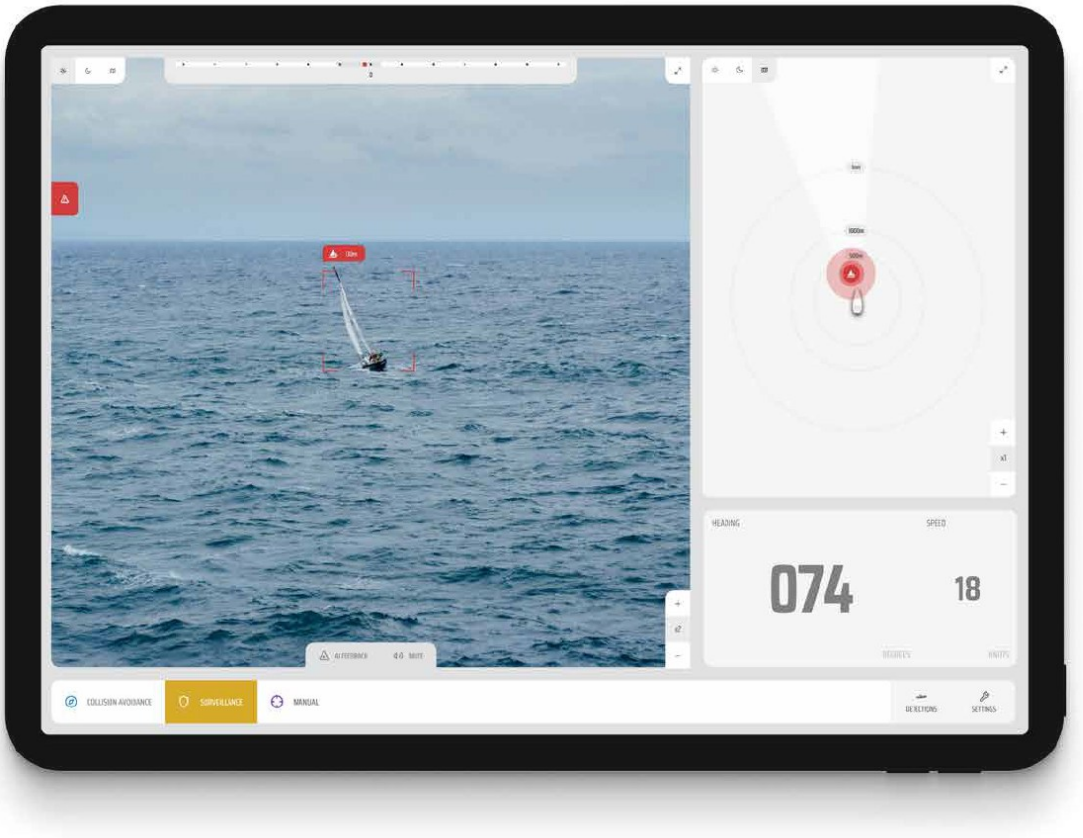
- Surveillance

▪In surveillance mode, the camera will perform a full 360° scan of the surroundings of the vessel. This mode is recommended when the vessel is not in motion.

▪You have the option to adjust the camera rotation speed in the "Personalize" tab in the settings section. You can choose between fast (30 seconds per 360°) and slow (110 seconds per 360°) rotation speeds.

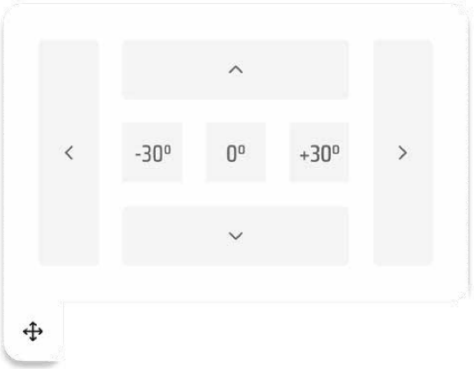
Please Note: In the fast mode, the detection range is reduced due to motion blur, and automatic detection is not performed on the long-range camera.

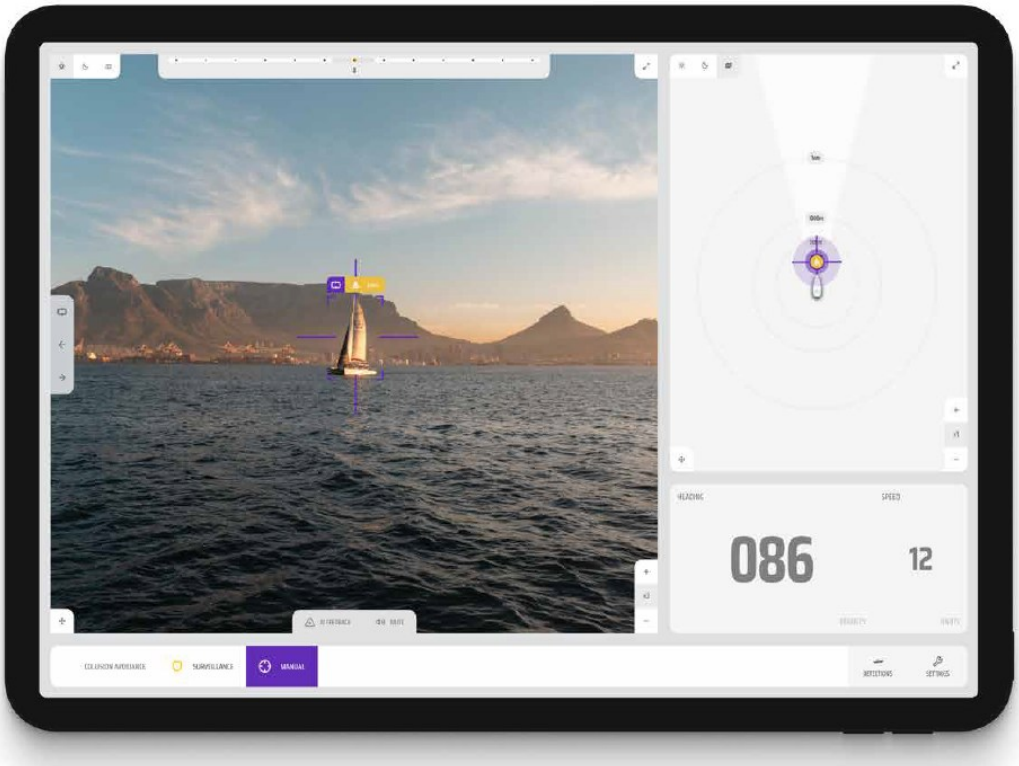




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- Camera control
  - Explore areas and objects of interest with ease. In camera control mode, you have full control over the camera's movements using the tilt and pan options available on the left panel buttons. The indication arrows allow you to rotate the camera, giving you comprehensive control over your viewing direction.





  - By pressing the +30 or -30 buttons, the camera will adjust it position by jumping 30° in the respective direction. Clicking in the 0° button will bring the camera back to its home position.

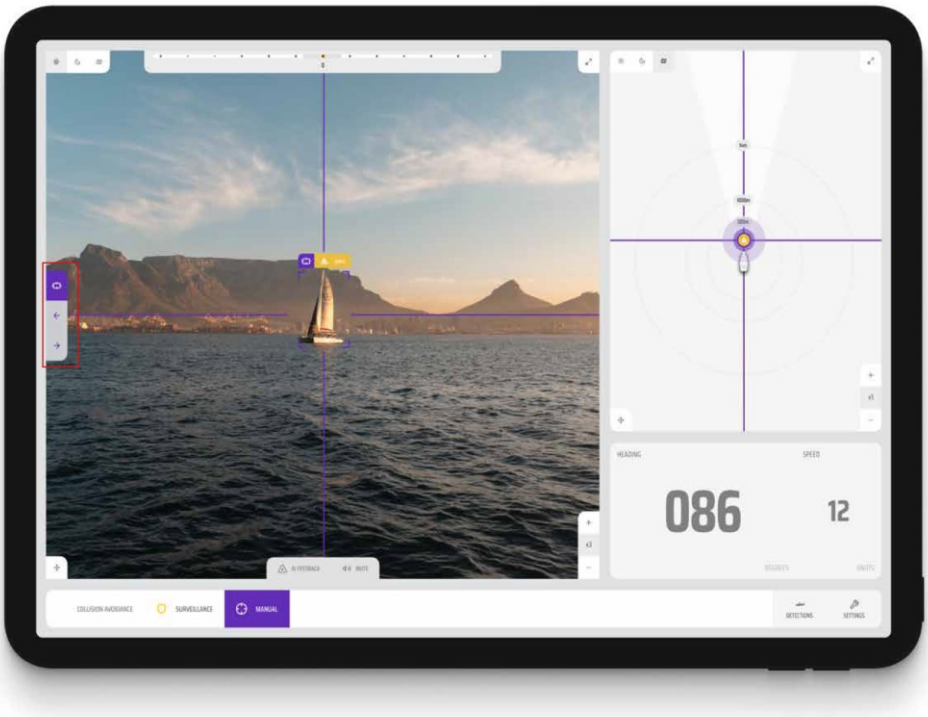
Please Note: The alarms function differently in each mode, catering to the specific features and needs of that mode. In camera control mode the sound alarm is disabled.

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- Tracking
  - In camera control mode, you also have the feature of tracking a detection. This feature allows you to follow and track detections, ensuring that you can easily capture the action of your objects of interest.
    This means the camera will intelligently follow and keep focus on an automatically selected detection, making it easier to capture the action of objects of interest as they move within the camera's field of view.
    To track an object follow these steps:
    - Begin with selecting the camera control mode.
    - Center/proximate the camera's crosshair on the desired detection. When the object is correctly aligned, it will be highlighted in purple, indicating it has been selected for tracking.



    - On the left side of the screen, you'll find the tracking menu. Locate and click on the "Track" button within this menu.



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- Upon clicking the "Track" button, a crosshair will become visible, featuring four complete lines spanning across the screen. Following this, the camera will operate intelligently by continuously tracking and retaining focus on the selected detection.

- Additionally, inside the tracking menu, you have the option to switch to targets left and right of the selected object. This allows you to choose through detected objects in case there are multiple candidates within the camera's field of view.

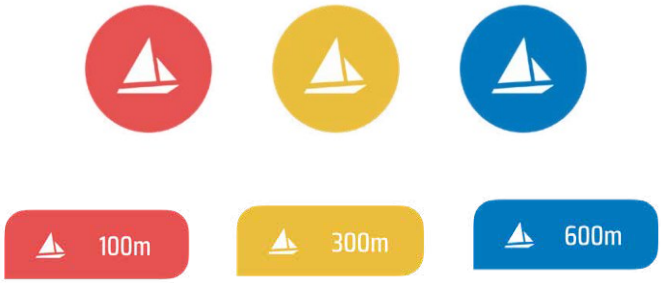
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5. Threat levels and Alarms

SEA.AI systems have three separate threat levels designed to provide users with optimum risk awareness depended on the operation mode. Each mode has a dedicated alarm structure.

- **Camera Control mode:** Only "blue" detections – no threat levels or alarms
- **Collision Avoidance:** Smart alarms – will automatically change between the 3 threat levels based on direction, range and size of the detection.
- **Surveillance Mode:** 3 configurable threat levels based on range

Each of the three threat levels includes colour-coded visual indications of detected objects.



In Collision avoidance and Surveillance mode, detected objects trigger audible warnings, alerts or even voice alarms.

- **Blue level** – *Detection:* No acoustic or visual alarm is generated.
- **Yellow level** – *Warning:* Triggers visual and acoustic warning
- **Red level** – *Alert:* Triggers visual and acoustic alert

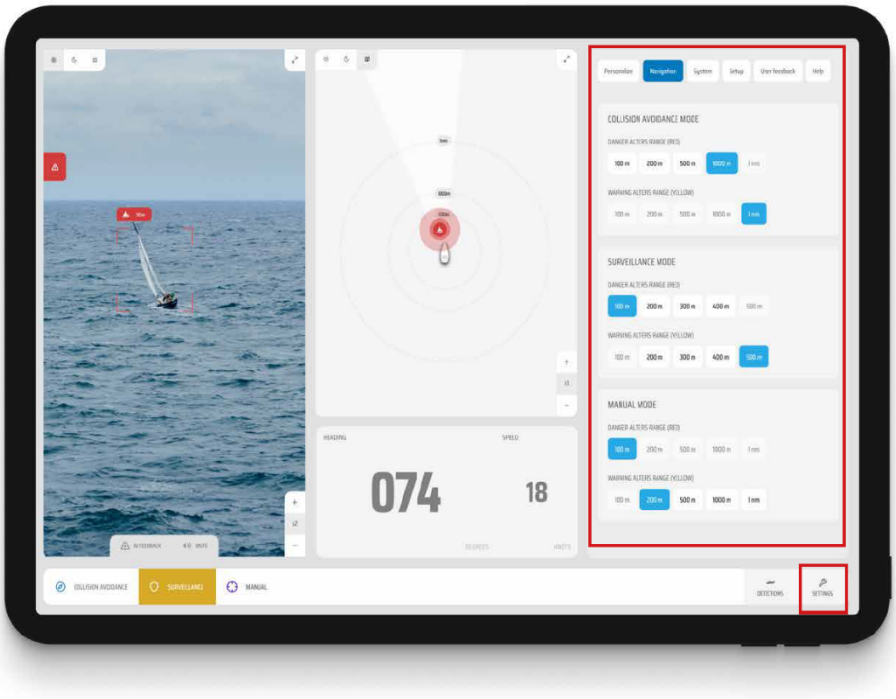
The *Voice Alarms* setting selects the alarm output type, enabling either acoustic tone alerts or synthesized voice alerts. This setting applies only to the audio on the SEA.AI APP; alarms generated by MFDs or external buzzers are not affected and will always emit acoustic beeps.

Please Note: Voice alarms can only be played via the SEA.AI App in combination with a sound speaker.

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Audible warnings can be easily muted using the sound icon.

You can also customize the distance parameters for each threat level to suit your individual needs within the navigation settings.



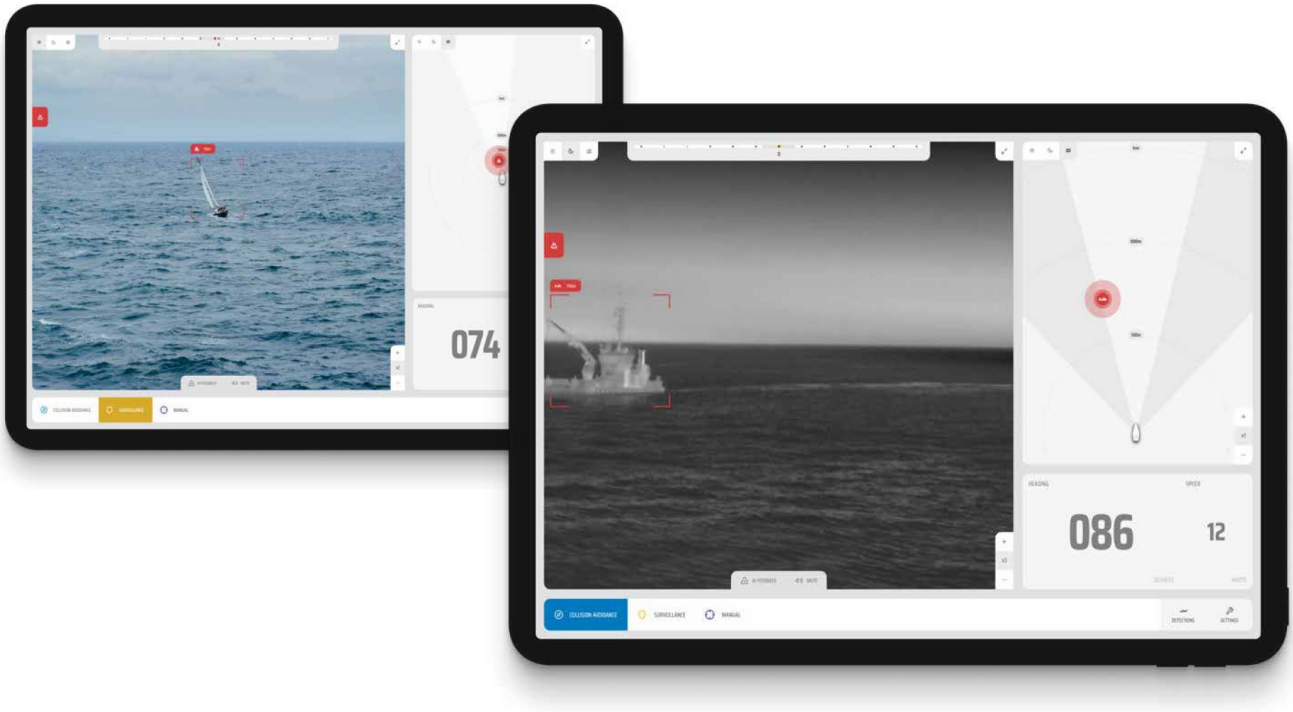
SEA.AI is an assistance system designed to enhance safety at sea. It does not replace the requirement for a proper lookout to be kept by the person in charge of the vessel.

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6. Detections

The SEA.AI system computes up to 10 images per second on both thermal sensors and displays live video images at a rate up to 24 frames per second.

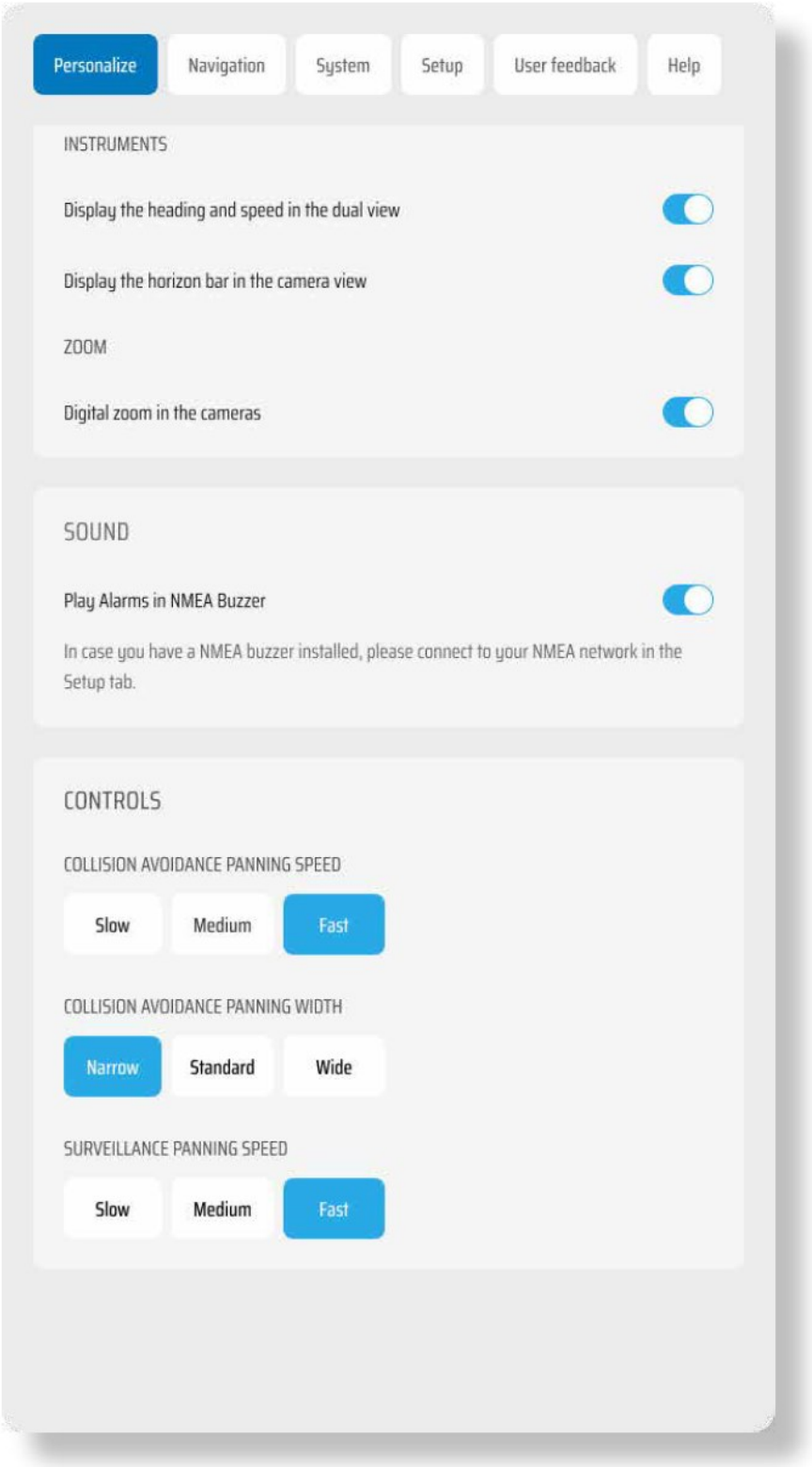
The detection and bounding boxes are effectively handled by both thermal cameras, ensuring reliable processing day and night, while the colour camera displays the overlaid bounding boxes for enhanced visual representation.



You can customize the alarm preferences, including threat levels and corresponding alarms in the settings. The threat levels and alarms are determined either by distance or in Collision Avoidance with smart alarms. The smart alarms will automatically change between the 3 threat levels based on direction, range and size of the detection. You have the option to adjust these parameters within the settings section.

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7. Setting



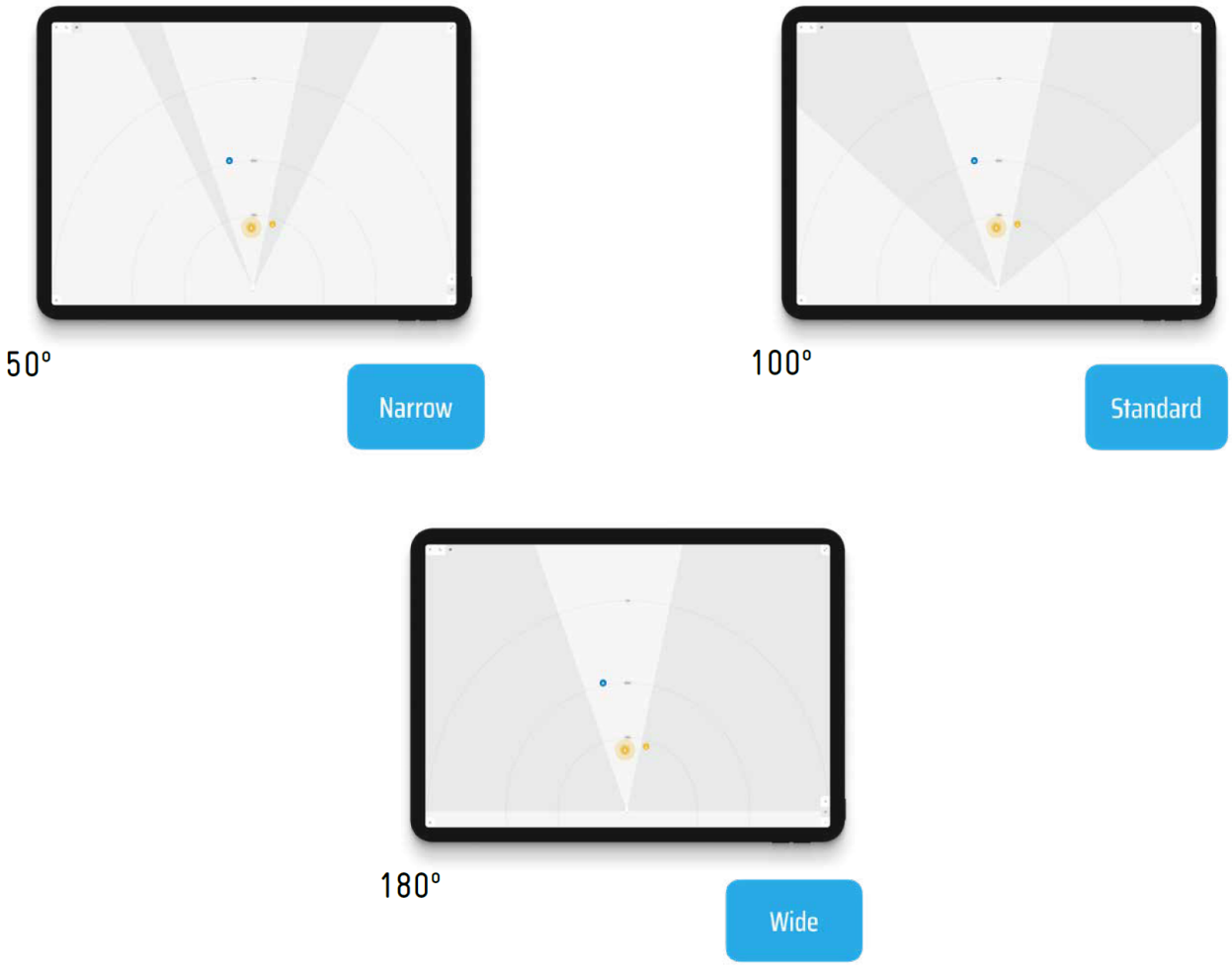
Personalize:

In the personalized settings, you have the option to adjust the panning width and speed for each operation mode according to your personal preferences. This means you can customize the camera's movement to cover the desired width and control the speed at which it pans.

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Panning width:

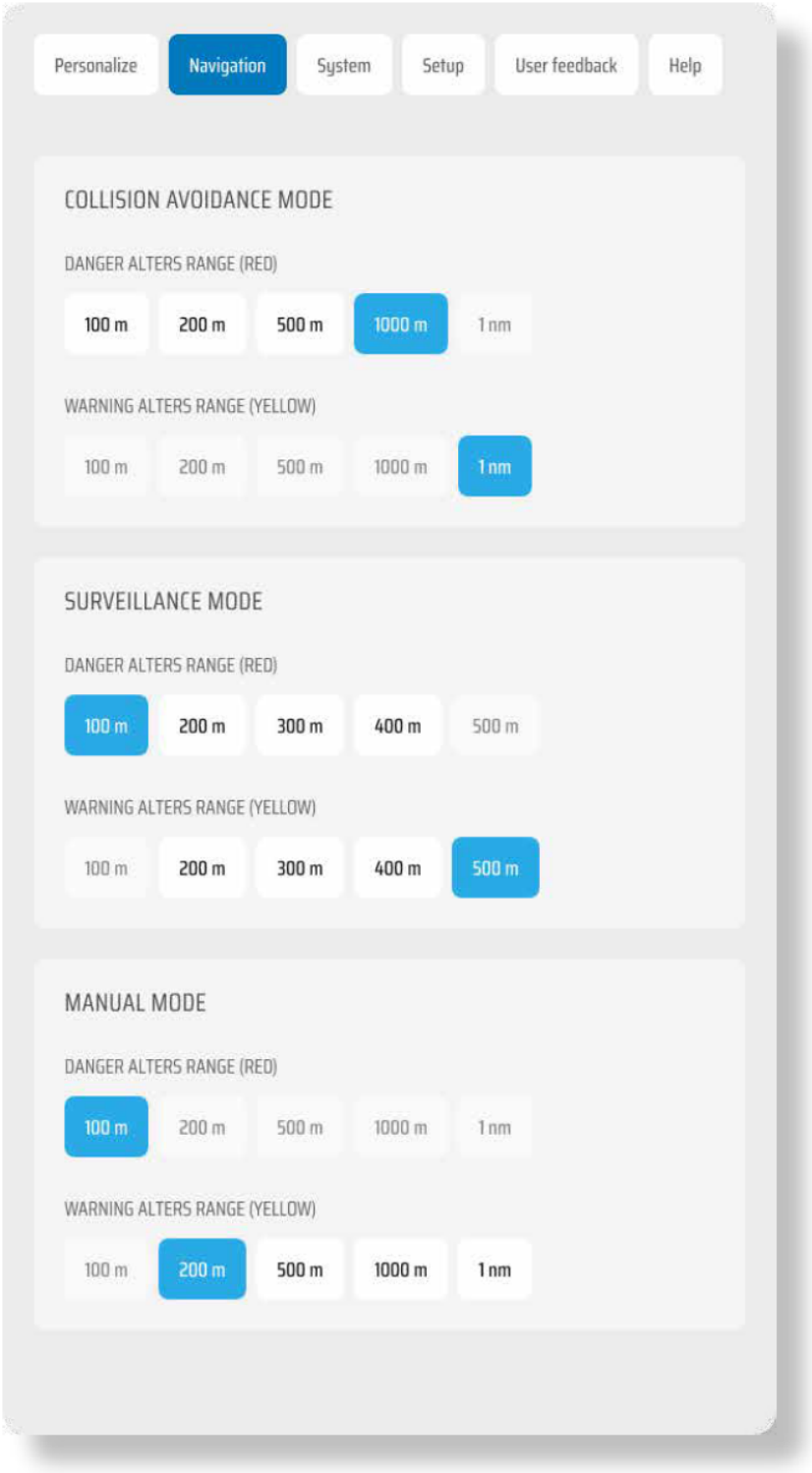
Choose from four different panning width options to customize the coverage based on your vessel and navigational environment: Forward view, Narrow, Standard or Wide.



Panning speed:

Adjust the panning speed to match the requirements of the situation. Select "fast" for dynamic scenarios and "slow" for precise monitoring during long-range detections. Tailoring these settings allows you to optimize your viewing experience and adapt it to your specific needs in different operation modes.

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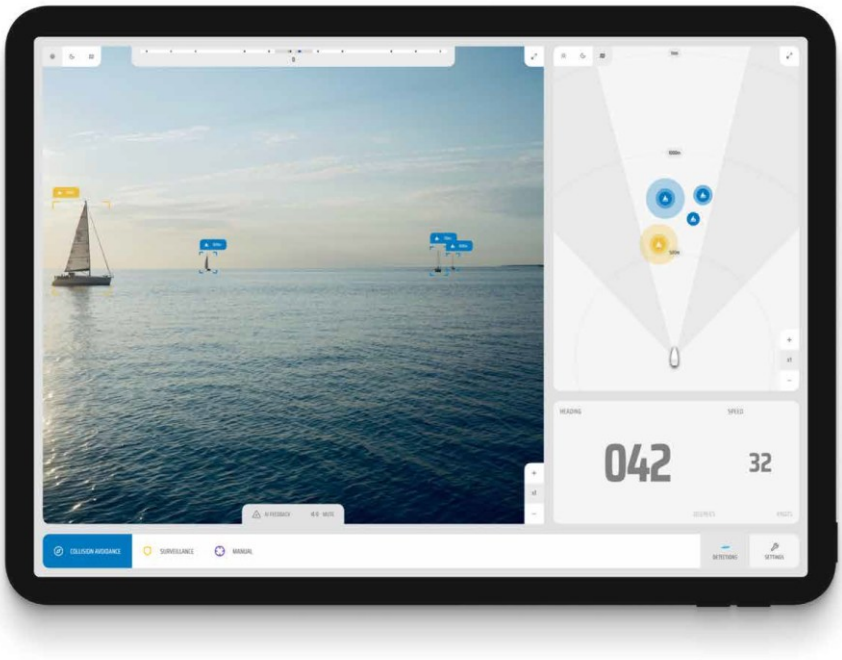


Navigation:

You have the ability to customize alarms for each operation mode, allowing you to select different warning and danger alert ranges.

This enables you to personalize the alarms based on your specific preferences and requirements by setting the range distances at which you want alarms to be triggered.

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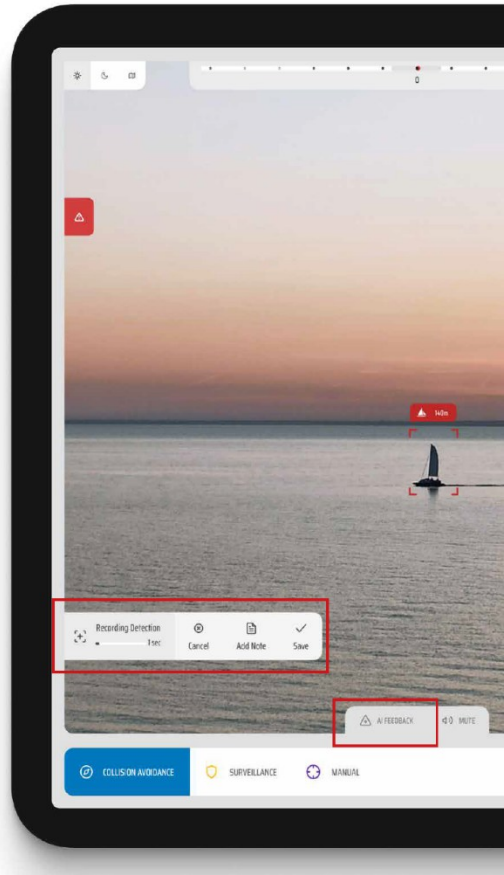
By configuring the alarms, you can ensure that you receive appropriate notifications and warnings during your navigation, enhancing your overall safety and awareness.

AI Feedback:

The AI Feedback feature allows you to record data and later send it to SEA.AI for system improvement.

By sharing your recordings, you play a crucial role in enhancing our AI, as this data helps us retrain and improve performance in edge cases.

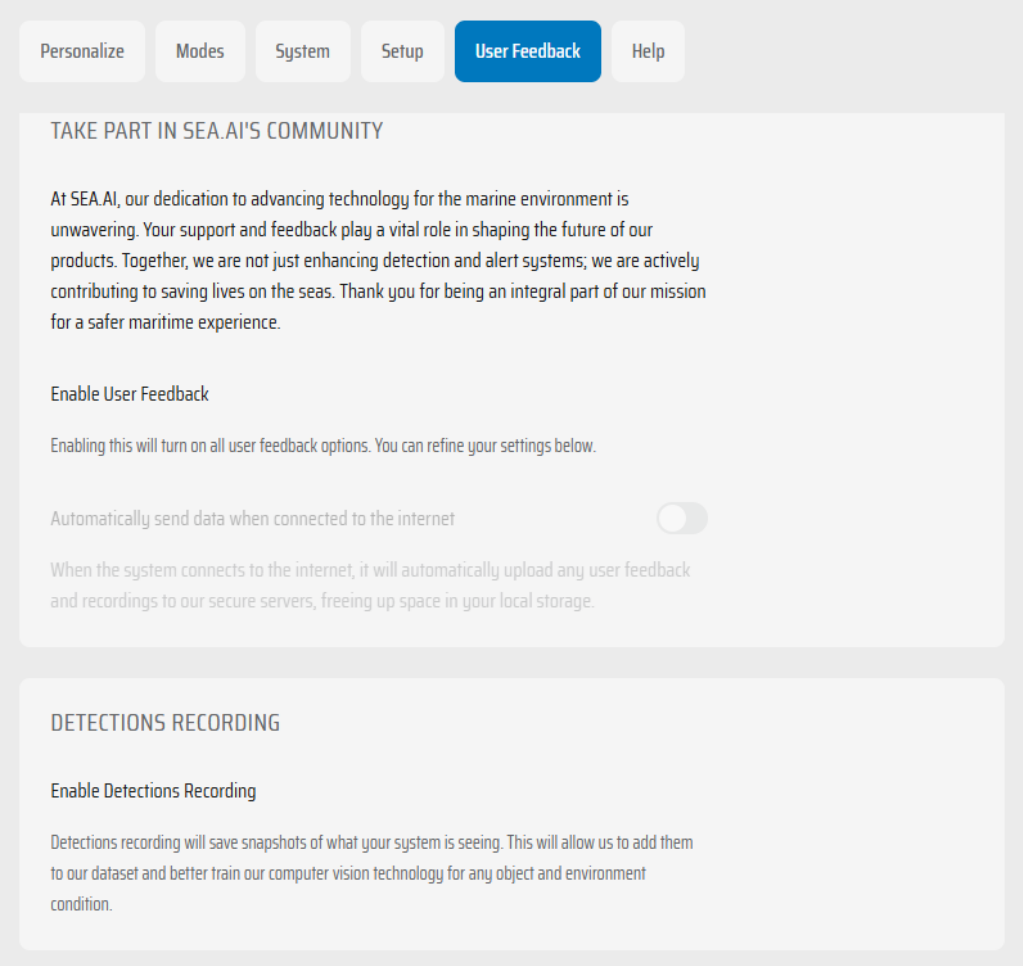
To send your recording you can reach out to us at: feedback@sea.ai



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Manage recordings:

To manage your recordings, navigate to the "Recordings" section in the settings menu. Here you can update, delete, or send recordings. Deleting older recordings can help free up storage space on your device.



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Advanced Settings:

Disclaimer *Advanced settings allow fine control over streaming performance and compatibility.*
These settings do not influence the AI and detection performance! However, incorrect values may cause video streams to malfunction or stop working on some devices e.g. MFDs.
*If you experience any issues after making changes, you can always restore the original configuration by clicking: "***Reset settings to default"**

This will safely revert all advanced options to their recommended values.

With the advanced settings you have control over image quality and bandwidth usage and thus over MFD compatibility. Higher resolution, frame rate, bit rate, and image quality improve visual performance however affect bandwidth and compatibility or performance with MFDs. If video performance issues occur, reduce resolution, frame rate, or bit rate.

SEA.AI's offers 3 layers of video/image output.

- **Video Streams** for the SEA.AI Applications (Mobile clients, PCs, and latest MFDs)
- **Image Streams** for the SEA.AI Application (used for clients who do not support Video streaming e.g. Garmin MFDs)
- **RTSP Streams** used for 3rd party integration e.g. recording SW or some MFDs

Accessing Advanced Settings

1. Open **Personalisation** in the application menu.
2. Enable **Advanced Settings**.
3. A new **Advanced** tab will appear in the Settings menu.
4. Open the **Advanced** tab to access streaming and camera configuration options.

Video Stream Settings

The Video Stream section controls how live video is transmitted from the camera.

Resolution

Users can select:

- **Low**
- **Standard**
- **High**

Note: Some multifunction displays (MFDs) may not support High resolution. If video does not display correctly, select a lower resolution.

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Encoder

Choose between:

- **H.264**
- **VP9**

Note: Not all devices support both encoders. If compatibility issues occur, switch to the alternative encoder.

Frame Rate

Controls the number of frames per second (FPS):

- Adjustable from **3 to 16 FPS**
- Default: **16 FPS**

Lower frame rates reduce bandwidth usage.

I-Frame Interval

The I-frame interval determines how frequently full reference frames are sent.

- Reducing the interval can improve image detail and stream stability.
- Lower values increase bandwidth usage.

Bit Rate

Bit rate controls bandwidth allocation:

- Adjustable from **1 to 6 Mbps**

Separate bit rate controls are available for:

- Color Wide Camera
- Color Narrow Camera
- Thermal Wide Camera
- Thermal Narrow Camera

Note: Higher bit rates improve image quality but require more network bandwidth.

Image Stream Settings

The Image Stream is used by browsers or systems that do not support video streaming.

Instead of continuous video, the system sends sequential images.

Available Settings

- **Resolution:** Low / Standard / High
- **Frame Rate:** 1 to 10 FPS

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- **Image Quality**: Replaces bit rate control

Higher image quality increases bandwidth usage.

RTSP Stream Settings

RTSP (Real-Time Streaming Protocol) is a standard protocol for integration

Aspect Ratio

Controls video cropping and visible field of view.

Detection Overlay

- **Enabled**: Detection boxes are drawn directly on the RTSP stream.
- **Disabled**: Stream shows clean live camera footage without overlays.

Resolution

- Low / Standard / High

Note: Some MFDs may not support High resolution as video Input.

Encoder

- **H.264** (currently the only available RTSP encoder)

Frame Rate

- Adjustable from **3 to 16 FPS**

I-Frame Interval

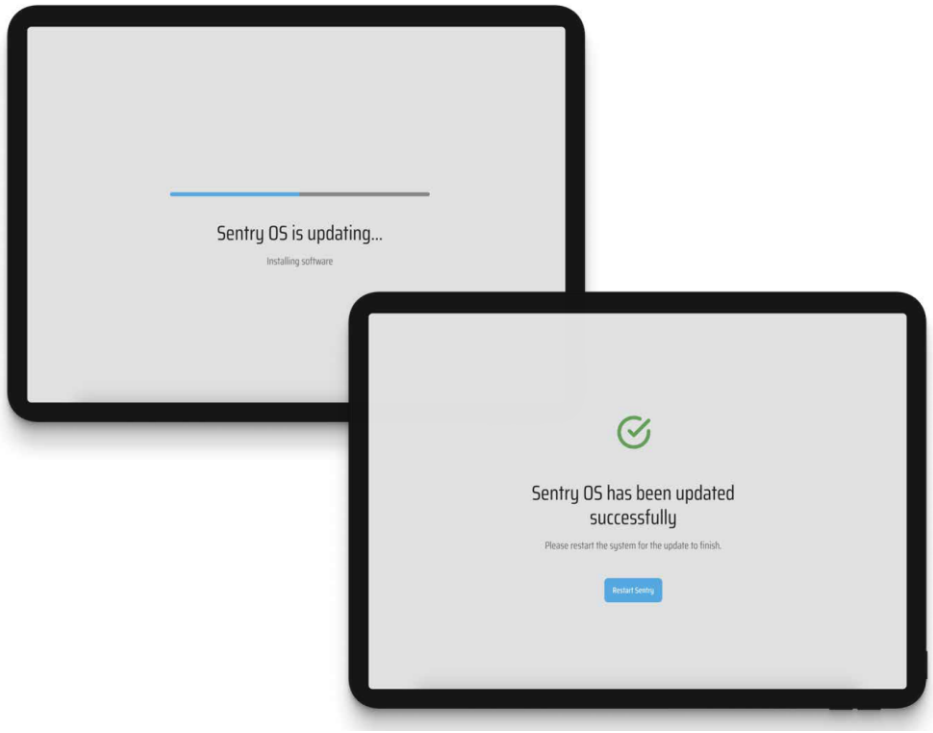
- Lower values improve image clarity
- Increases bandwidth consumption

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8. Updates

Stay up to date by keeping your system regularly updated.

You will receive new updates in your registered email, ensuring future improvements and enhancements for your system.



Here's how to update your system:

1. Access the link sent to your registered email or directly through the provided link below.
2. Open the folder "Software updates".
3. Download the update package (oscar file).
4. Start Sentry and open the App.
5. Go to the setting menu.
6. Select "System".
7. Choose "System update" and click on "Upload File".
8. Upload the package file that you downloaded from the "Software updates" folder.

Direct link to Software Updates folder

If you encounter any problems with this procedure, please contact your local dealer/installer or reach out to SEA.AI directly via email at: service@sea.ai

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9. Service and support



Over the air updates and performance improvements:

SEA.AI's artificial intelligence engine is based around an ever-expanding database which we update at regular intervals making SEA.AI even more efficient. If you need any help with updating, please check the update section of this guide, or contact us for further help.



Remote software and service support:

SEA.AI can provide remote support to ensure the reliable performance of your system wherever in the world our users' vessels are operating. Contact us for online remote support.



Feedback:

If you have any thoughts or comments about our products or customer service we welcome your feedback via: feedback@sea.ai



Contact us:

To contact us, please use our email address service@sea.ai and we will be happy to assist you. You can also find these contact details in the help section of your settings in the app.

SEA.AI

www.sea.ai

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